2.28 Integration: Substitutions Mix

Worked Solutions

1 a)

$$\frac{1}{\sqrt{a^2-x^2}} = \frac{1}{\sqrt{a^2-a^2\sin^2\theta}} = \frac{1}{\sqrt{a^2(1-\sin^2\theta)}}$$
$$= \frac{1}{\sqrt{a^2\cos^2\theta}} = \frac{1}{a\cos\theta}.$$

So use $x = a\sin\theta$.

1 b)

$$\frac{1}{\sqrt{a^2+x^2}} = \frac{1}{\sqrt{a^2+a^2\sinh^2\theta}} = \frac{1}{\sqrt{a^2(1+\sinh^2\theta)}}$$
$$= \frac{1}{\sqrt{a^2\cosh^2\theta}} = \frac{1}{a\cosh\theta}.$$

So use $x = a\sinh\theta$.

1 c)

$$\frac{1}{a^2-x^2} = \frac{1}{a^2-a^2\tanh^2\theta} = \frac{1}{a^2(1-\tanh^2\theta)} = \frac{1}{a^2\,\mathrm{sech}^2\,\theta}.$$

So use $x = a\tanh\theta$.

1 d)

$$\frac{1}{\sqrt{x^2-a^2}} = \frac{1}{\sqrt{a^2\cosh^2\theta-a^2}} = \frac{1}{\sqrt{a^2(\cosh^2\theta-1)}}$$
$$= \frac{1}{\sqrt{a^2\sinh^2\theta}} = \frac{1}{a\sinh\theta}.$$

So use $x = a\cosh\theta$.

1 e)

$$\frac{1}{a^2+x^2} = \frac{1}{a^2+a^2\tan^2\theta} = \frac{1}{a^2(1+\tan^2\theta)} = \frac{1}{a^2\sec^2\theta}.$$

So use $x = a\tan\theta$.

2 a)

Let $9x^2 = 400\sinh^2 u$, so $3x = 20\sinh u$ and $x = \frac{20}{3}\sinh u$. Then $dx = \frac{20}{3}\cosh u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{9x^2+400}}\,dx &= \int \frac{1}{\sqrt{400\sinh^2 u+400}}\cdot\frac{20}{3}\cosh u\,du\\ &= \int \frac{1}{\sqrt{400(\sinh^2 u+1)}}\cdot\frac{20}{3}\cosh u\,du\\ &= \frac{1}{3}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du = \frac{1}{3}\int 1\,du\\ &= \frac{1}{3}u + c = \frac{1}{3}\operatorname{arsinh}\left(\frac{3x}{20}\right)+c.\end{aligned}$$

So $x = \frac{20}{3}\sinh u$ gives $\int \frac{1}{\sqrt{9x^2+400}}\,dx = \frac{1}{3}\operatorname{arsinh}\left(\frac{3x}{20}\right)+c$.

2 b)

Let $16x^2 = 9\sin^2 u$, so $4x = 3\sin u$ and $x = \frac{3}{4}\sin u$. Then $dx = \frac{3}{4}\cos u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{9-16x^2}}\,dx &= \int \frac{1}{\sqrt{9-9\sin^2 u}}\cdot\frac{3}{4}\cos u\,du\\ &= \int \frac{1}{\sqrt{9(1-\sin^2 u)}}\cdot\frac{3}{4}\cos u\,du = \frac{1}{4}\int \frac{\cos u}{\sqrt{\cos^2 u}}\,du\\ &= \frac{1}{4}\int 1\,du = \frac{1}{4}u + c = \frac{1}{4}\arcsin\left(\frac{4x}{3}\right)+c.\end{aligned}$$

So $x = \frac{3}{4}\sin u$ gives $\int \frac{1}{\sqrt{9-16x^2}}\,dx = \frac{1}{4}\arcsin\left(\frac{4x}{3}\right)+c$.

2 c)

Let $x^2 = 36\tanh^2 u$, so $x = 6\tanh u$ and $dx = 6\operatorname{sech}^2 u\,du$.

$$\begin{aligned}\int \frac{1}{36-x^2}\,dx &= \int \frac{1}{36-36\tanh^2 u}\,6\operatorname{sech}^2 u\,du\\ &= \int \frac{1}{36(1-\tanh^2 u)}\,6\operatorname{sech}^2 u\,du = \frac{1}{6}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,du\\ &= \frac{1}{6}\int 1\,du = \frac{1}{6}u + c = \frac{1}{6}\operatorname{artanh}\left(\frac{1}{6x}\right)+c.\end{aligned}$$

So $x = 6\tanh u$ gives $\int \frac{1}{36-x^2}\,dx = \frac{1}{6}\operatorname{artanh}\left(\frac{x}{6}\right)+c$.

2 d)

Let $625x^2 = \sinh^2 u$, so $25x = \sinh u$ and $x = \frac{1}{25}\sinh u$. Then $dx = \frac{1}{25}\cosh u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{625x^2+1}}\,dx &= \int \frac{1}{\sqrt{\sinh^2 u+1}}\cdot\frac{1}{25}\cosh u\,du\\ &= \frac{1}{25}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du = \frac{1}{25}\int 1\,du\\ &= \frac{1}{25}u + c = \frac{1}{25}\operatorname{arsinh}(25x)+c.\end{aligned}$$

So $x = \frac{1}{25}\sinh u$ gives $\int \frac{1}{\sqrt{625x^2+1}}\,dx = \frac{1}{25}\operatorname{arsinh}(25x)+c$.

2 e)

Let $121x^2 = 16\cosh^2 u$, so $11x = 4\cosh u$ and $x = \frac{4}{11}\cosh u$. Then $dx = \frac{4}{11}\sinh u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{121x^2-16}}\,dx &= \int \frac{1}{\sqrt{16\cosh^2 u - 16}}\cdot\frac{4}{11}\sinh u\,du \\ &= \int \frac{1}{\sqrt{16(\cosh^2 u - 1)}}\cdot\frac{4}{11}\sinh u\,du = \frac{1}{11}\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\,du \\ &= \frac{1}{11}\int 1\,du = \frac{1}{11}u + c = \frac{1}{11}\operatorname{arcosh}\left(\frac{11x}{4}\right) + c.\end{aligned}$$

So $x = \frac{4}{11}\cosh u$ gives $\int \frac{1}{\sqrt{121x^2-16}}\,dx = \frac{1}{11}\operatorname{arcosh}\left(\frac{11x}{4}\right) + c$.

2 f)

Let $25x^2 = \tan^2 u$, so $5x = \tan u$ and $x = \frac{1}{5}\tan u$. Then $dx = \frac{1}{5}\sec^2 u\,du$.

$$\begin{aligned}\int \frac{1}{1+25x^2}\,dx &= \int \frac{1}{1+\tan^2 u}\cdot\frac{1}{5}\sec^2 u\,du = \frac{1}{5}\int \frac{\sec^2 u}{\sec^2 u}\,du \\ &= \frac{1}{5}\int 1\,du = \frac{1}{5}u + c = \frac{1}{5}\arctan(5x) + c.\end{aligned}$$

So $x = \frac{1}{5}\tan u$ gives $\int \frac{1}{1+25x^2}\,dx = \frac{1}{5}\arctan(5x) + c$.

2 g)

Let $100x^2 = 9\sinh^2 u$, so $10x = 3\sinh u$ and $x = \frac{3}{10}\sinh u$. Then $dx = \frac{3}{10}\cosh u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{100x^2+9}}\,dx &= \int \frac{1}{\sqrt{9\sinh^2 u + 9}}\cdot\frac{3}{10}\cosh u\,du \\ &= \int \frac{1}{\sqrt{9(\sinh^2 u + 1)}}\cdot\frac{3}{10}\cosh u\,du = \frac{1}{10}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du \\ &= \frac{1}{10}\int 1\,du = \frac{1}{10}u + c = \frac{1}{10}\operatorname{arsinh}\left(\frac{10x}{3}\right) + c.\end{aligned}$$

So $x = \frac{3}{10}\sinh u$ gives $\int \frac{1}{\sqrt{100x^2+9}}\,dx = \frac{1}{10}\operatorname{arsinh}\left(\frac{10x}{3}\right) + c$.

2 h)

Let $121x^2 = 4\sin^2 u$, so $11x = 2\sin u$ and $x = \frac{2}{11}\sin u$. Then $dx = \frac{2}{11}\cos u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{4-121x^2}}\,dx &= \int \frac{1}{\sqrt{4-4\sin^2 u}}\cdot\frac{2}{11}\cos u\,du \\ &= \int \frac{1}{\sqrt{4(1-\sin^2 u)}}\cdot\frac{2}{11}\cos u\,du = \frac{1}{11}\int \frac{\cos u}{\sqrt{\cos^2 u}}\,du \\ &= \frac{1}{11}\int 1\,du = \frac{1}{11}u + c = \frac{1}{11}\arcsin\left(\frac{11x}{2}\right) + c.\end{aligned}$$

So $x = \frac{2}{11}\sin u$ gives $\int \frac{1}{\sqrt{4-121x^2}}\,dx = \frac{1}{11}\arcsin\left(\frac{11x}{2}\right) + c$.

2 i)

Let $x^2 = 4\cosh^2 u$, so $x = 2\cosh u$ and $dx = 2\sinh u\, du$.

$$\begin{aligned}\int \frac{16}{\sqrt{x^2-4}}\, dx &= \int \frac{16}{\sqrt{4\cosh^2 u - 4}}\, 2\sinh u\, du \\ &= 16\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\, du = 16\int 1\, du \\ &= 16u + c = 16\operatorname{arcosh}\left(\frac{x}{2}\right) + c.\end{aligned}$$

So $x = 2\cosh u$ gives $\displaystyle\int \frac{16}{\sqrt{x^2-4}}\, dx = 16\operatorname{arcosh}\left(\frac{x}{2}\right) + c$.

2 j)

Let $49x^2 = 144\tan^2 u$, so $7x = 12\tan u$ and $x = \frac{12}{7}\tan u$. Then $dx = \frac{12}{7}\sec^2 u\, du$.

$$\begin{aligned}\int \frac{7}{144+49x^2}\, dx &= \int \frac{7}{144+144\tan^2 u}\cdot\frac{12}{7}\sec^2 u\, du \\ &= \frac{1}{12}\int \frac{\sec^2 u}{\sec^2 u}\, du = \frac{1}{12}u + c = \frac{1}{12}\arctan\left(\frac{7x}{12}\right) + c.\end{aligned}$$

So $x = \frac{12}{7}\tan u$ gives $\displaystyle\int \frac{7}{144+49x^2}\, dx = \frac{1}{12}\arctan\left(\frac{7x}{12}\right) + c$.

2 k)

Let $9x^2 = 25\tanh^2 u$, so $3x = 5\tanh u$ and $x = \frac{5}{3}\tanh u$. Then $dx = \frac{5}{3}\operatorname{sech}^2 u\, du$.

$$\begin{aligned}\int \frac{24}{25-9x^2}\, dx &= \int \frac{24}{25-25\tanh^2 u}\cdot\frac{5}{3}\operatorname{sech}^2 u\, du \\ &= \frac{8}{5}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\, du = \frac{8}{5}u + c = \frac{8}{5}\operatorname{artanh}\left(\frac{3x}{5}\right) + c.\end{aligned}$$

So $x = \frac{5}{3}\tanh u$ gives $\displaystyle\int \frac{24}{25-9x^2}\, dx = \frac{8}{5}\operatorname{artanh}\left(\frac{3x}{5}\right) + c$.

2 l)

Let $2x^2 = 81\sin^2 u$, so $\sqrt{2}x = 9\sin u$ and $x = \frac{9}{\sqrt{2}}\sin u$. Then $dx = \frac{9\sqrt{2}}{2}\cos u\, du$.

$$\begin{aligned}\int \frac{32}{\sqrt{81-2x^2}}\, dx &= \int \frac{32}{\sqrt{81-81\sin^2 u}}\cdot\frac{9\sqrt{2}}{2}\cos u\, du \\ &= 16\sqrt{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\, du = 16\sqrt{2}u + c \\ &= 16\sqrt{2}\arcsin\left(\frac{\sqrt{2}x}{9}\right) + c.\end{aligned}$$

So $x = \frac{9}{\sqrt{2}}\sin u$ gives $\displaystyle\int \frac{32}{\sqrt{81-2x^2}}\, dx = 16\sqrt{2}\arcsin\left(\frac{\sqrt{2}x}{9}\right) + c$.

2 m)

Let $9x^2 = 10\cosh^2 u$, so $3x = \sqrt{10}\cosh u$ and $x = \frac{\sqrt{10}}{3}\cosh u$. Then $dx = \frac{\sqrt{10}}{3}\sinh u\, du$.

$$\int \frac{90}{\sqrt{9x^2-10}}\, dx = \int \frac{90}{\sqrt{10\cosh^2 u - 10}}\cdot\frac{\sqrt{10}}{3}\sinh u\, du$$

$$= 30 \int \frac{\sinh u}{\sqrt{\sinh^2 u}}\, du = 30u + c$$
$$= 30 \operatorname{arcosh}\left(\frac{3x}{\sqrt{10}}\right) + c = 30 \operatorname{arcosh}\left(\frac{3\sqrt{10}x}{10}\right) + c.$$

So $x = \frac{\sqrt{10}}{3}\cosh u$ gives $\displaystyle\int \frac{90}{\sqrt{9x^2-10}}\, dx = 30 \operatorname{arcosh}\left(\frac{3\sqrt{10}x}{10}\right) + c.$

2 n)

Let $121x^2 = 50\tanh^2 u$, so $11x = \sqrt{50}\tanh u$ and $x = \frac{\sqrt{50}}{11}\tanh u$. Then $dx = \frac{5\sqrt{2}}{11}\operatorname{sech}^2 u\, du$.

$$\int \frac{55}{50-121x^2}\, dx = \int \frac{55}{50-50\tanh^2 u} \cdot \frac{5\sqrt{2}}{11}\operatorname{sech}^2 u\, du$$
$$= \frac{\sqrt{2}}{2}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\, du = \frac{\sqrt{2}}{2}u + c$$
$$= \frac{\sqrt{2}}{2}\operatorname{artanh}\left(\frac{11x}{\sqrt{50}}\right) + c = \frac{\sqrt{2}}{2}\operatorname{artanh}\left(\frac{11\sqrt{2}x}{10}\right) + c.$$

So $x = \frac{\sqrt{50}}{11}\tanh u$ gives $\displaystyle\int \frac{55}{50-121x^2}\, dx = \frac{\sqrt{2}}{2}\operatorname{artanh}\left(\frac{11\sqrt{2}x}{10}\right) + c.$

2 o)

Let $108x^2 = 5\tan^2 u$, so $6\sqrt{3}x = \sqrt{5}\tan u$ and $x = \frac{\sqrt{5}}{6\sqrt{3}}\tan u$. Then $dx = \frac{\sqrt{15}}{18}\sec^2 u\, du$.

$$\int \frac{36}{5+108x^2}\, dx = \int \frac{36}{5+5\tan^2 u} \cdot \frac{\sqrt{15}}{18}\sec^2 u\, du$$
$$= \frac{2\sqrt{15}}{5}\int \frac{\sec^2 u}{\sec^2 u}\, du = \frac{2\sqrt{15}}{5}u + c$$
$$= \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{3}x}{\sqrt{5}}\right) + c = \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{15}x}{5}\right) + c.$$

So $x = \frac{\sqrt{5}}{6\sqrt{3}}\tan u$ gives $\displaystyle\int \frac{36}{5+108x^2}\, dx = \frac{2\sqrt{15}}{5}\arctan\left(\frac{6\sqrt{15}x}{5}\right) + c.$

3 a)

Let $9x^2 = 4\sinh^2 u$, so $3x = 2\sinh u$ and $x = \frac{2}{3}\sinh u$. Then $dx = \frac{2}{3}\cosh u\, du$.

$$\int \frac{3}{\sqrt{9x^2+4}}\, dx = \int \frac{3}{\sqrt{4\sinh^2 u + 4}} \cdot \frac{2}{3}\cosh u\, du = \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\, du$$
$$= \int 1\, du = u + c = \operatorname{arsinh}\left(\frac{3x}{2}\right) + c.$$

So $x = \frac{2}{3}\sinh u$ gives $\displaystyle\int \frac{3}{\sqrt{9x^2+4}}\, dx = \operatorname{arsinh}\left(\frac{3x}{2}\right) + c.$

3 b)

Let $49x^2 = 4\tan^2 u$, so $7x = 2\tan u$ and $x = \frac{2}{7}\tan u$. Then $dx = \frac{2}{7}\sec^2 u\, du$.

$$\int \frac{14}{4+49x^2}\, dx = \int \frac{14}{4+4\tan^2 u} \cdot \frac{2}{7}\sec^2 u\, du = \int \frac{\sec^2 u}{\sec^2 u}\, du$$

$$= \int 1\,du = u + c = \arctan\left(\frac{7x}{2}\right) + c.$$

So $x = \frac{2}{7}\tan u$ gives $\displaystyle\int \frac{14}{4+49x^2}\,dx = \arctan\left(\frac{7x}{2}\right) + c.$

3 c)

Let $16x^2 = 9\tanh^2 u$, so $4x = 3\tanh u$ and $x = \frac{3}{4}\tanh u$. Then $dx = \frac{3}{4}\operatorname{sech}^2 u\,du$.

$$\int \frac{144}{9-16x^2}\,dx = \int \frac{144}{9-9\tanh^2 u}\cdot\frac{3}{4}\operatorname{sech}^2 u\,du = 12\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\,du$$
$$= 12u + c = 12\operatorname{artanh}\left(\frac{4x}{3}\right) + c.$$

So $x = \frac{3}{4}\tanh u$ gives $\displaystyle\int \frac{144}{9-16x^2}\,dx = 12\operatorname{artanh}\left(\frac{4x}{3}\right) + c.$

3 d)

Let $16x^2 = 9\cosh^2 u$, so $4x = 3\cosh u$ and $x = \frac{3}{4}\cosh u$. Then $dx = \frac{3}{4}\sinh u\,du$.

$$\int \frac{8}{\sqrt{16x^2-9}}\,dx = \int \frac{8}{\sqrt{9\cosh^2 u - 9}}\cdot\frac{3}{4}\sinh u\,du = 2\int \frac{\sinh u}{\sqrt{\sinh^2 u}}\,du$$
$$= 2\int 1\,du = 2u + c = 2\operatorname{arcosh}\left(\frac{4x}{3}\right) + c.$$

So $x = \frac{3}{4}\cosh u$ gives $\displaystyle\int \frac{8}{\sqrt{16x^2-9}}\,dx = 2\operatorname{arcosh}\left(\frac{4x}{3}\right) + c.$

3 e)

Let $4x^2 = 25\sin^2 u$, so $2x = 5\sin u$ and $x = \frac{5}{2}\sin u$. Then $dx = \frac{5}{2}\cos u\,du$.

$$I = \int \sqrt{25-4x^2}\,dx = \int \sqrt{25-25\sin^2 u}\cdot\frac{5}{2}\cos u\,du = \frac{25}{2}\int \cos^2 u\,du.$$

Using $\cos^2 A = \frac{1}{2}\cos 2A + \frac{1}{2}$,

$$\begin{aligned} I &= \frac{25}{2}\int\left(\frac{1}{2}\cos 2u + \frac{1}{2}\right)du = \frac{25}{8}\sin 2u + \frac{25}{4}u + c \\ &= \frac{25}{4}\sin u\cos u + \frac{25}{4}u + c \\ &= \frac{25}{4}\sin u\sqrt{1-\sin^2 u} + \frac{25}{4}u + c \\ &= \frac{5}{2}x\sqrt{1-\frac{4x^2}{25}} + \frac{25}{4}\arcsin\left(\frac{2x}{5}\right) + c \\ &= \frac{1}{2}x\sqrt{25-4x^2} + \frac{25}{4}\arcsin\left(\frac{2x}{5}\right) + c. \end{aligned}$$

So $x = \frac{5}{2}\sin u$ gives $\displaystyle\int \sqrt{25-4x^2}\,dx = \frac{1}{2}x\sqrt{25-4x^2} + \frac{25}{4}\arcsin\left(\frac{2x}{5}\right) + c.$

3 f)

Let $9x^2 = 16\tanh^2 u$, so $3x = 4\tanh u$ and $x = \frac{4}{3}\tanh u$. Then $dx = \frac{4}{3}\operatorname{sech}^2 u\,du$.

$$I = \int \frac{24576}{(16-9x^2)^3}\,dx = \int \frac{24576}{(16-16\tanh^2 u)^3}\cdot\frac{4}{3}\operatorname{sech}^2 u\,du$$

$$= 8\int \frac{\operatorname{sech}^2 u}{(\operatorname{sech}^2 u)^3}\,du = 8\int \cosh^4 u\,du.$$

Since $\cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$,

$$\cosh^4 u = \left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)^2 = \frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8},$$
$$I = 8\int \left(\frac{1}{8}\cosh 4u + \frac{1}{2}\cosh 2u + \frac{3}{8}\right) du$$
$$= \frac{1}{4}\sinh 4u + 2\sinh 2u + 3u + c$$
$$= 2\sinh u\cosh^3 u + 3\sinh u\cosh u + 3u + c.$$

If $\tanh u = \frac{3}{4}x$, then $\cosh^2 u = \frac{16}{16-9x^2}$ and $\sinh u = \frac{3x}{\sqrt{16-9x^2}}$. Therefore

$$I = 2\frac{3x}{\sqrt{16-9x^2}}\left(\frac{4}{\sqrt{16-9x^2}}\right)^3 + 3\frac{3x}{\sqrt{16-9x^2}}\frac{4}{\sqrt{16-9x^2}} + 3\operatorname{artanh}\left(\frac{3x}{4}\right) + c$$
$$= \frac{384x}{(16-9x^2)^2} + \frac{36x}{16-9x^2} + 3\operatorname{artanh}\left(\frac{3x}{4}\right) + c$$
$$= \frac{960x - 324x^3}{(16-9x^2)^2} + 3\operatorname{artanh}\left(\frac{3x}{4}\right) + c.$$

So $x = \frac{4}{3}\tanh u$ gives $\displaystyle\int \frac{24576}{(16-9x^2)^3}\,dx = \frac{960x-324x^3}{(16-9x^2)^2} + 3\operatorname{artanh}\left(\frac{3x}{4}\right) + c.$

3 g)

Let $49x^2 = 4\sin^2 u$, so $7x = 2\sin u$ and $x = \frac{2}{7}\sin u$. Then $dx = \frac{2}{7}\cos u\,du$.

$$I = \int (4-49x^2)^{3/2}\,dx = \frac{16}{7}\int \cos^4 u\,du.$$

Using $\cos^4 u = \frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}$,

$$I = \frac{16}{7}\int \left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right) du$$
$$= \frac{1}{14}\sin 4u + \frac{4}{7}\sin 2u + \frac{6}{7}u + c$$
$$= \frac{4}{7}\sin u\cos^3 u + \frac{6}{7}\sin u\cos u + \frac{6}{7}u + c.$$

If $\sin u = \frac{7}{2}x$, then $\cos u = \frac{1}{2}\sqrt{4-49x^2}$. Hence

$$I = \frac{1}{4}x(4-49x^2)^{3/2} + \frac{3}{2}x\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7x}{2}\right) + c$$
$$= \frac{1}{4}x(10-49x^2)\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7x}{2}\right) + c.$$

So $x = \frac{2}{7}\sin u$ gives $\displaystyle\int \frac{14}{4+49x^2}\,dx = \frac{1}{4}x(10-49x^2)\sqrt{4-49x^2} + \frac{6}{7}\arcsin\left(\frac{7x}{2}\right) + c.$

3 h)

Let $36x^2 = 25\sinh^2 u$, so $6x = 5\sinh u$ and $x = \frac{5}{6}\sinh u$. Then $dx = \frac{5}{6}\cosh u\,du$.

$$I = \int \sqrt{36x^2+25}\,dx = \frac{25}{6}\int \cosh^2 u\,du$$

$$= \frac{25}{6}\int\left(\frac{1}{2}\cosh 2u + \frac{1}{2}\right)du = \frac{25}{24}\sinh 2u + \frac{25}{12}u + c$$
$$= \frac{25}{12}\sinh u\cosh u + \frac{25}{12}u + c$$
$$= \frac{1}{2}x\sqrt{36x^2+25} + \frac{25}{12}\operatorname{arsinh}\left(\frac{6x}{5}\right) + c.$$

So $x = \frac{5}{6}\sinh u$ gives the stated integral.

3 i)

Let $9x^2 = 100\tan^2 u$, so $3x = 10\tan u$ and $x = \frac{10}{3}\tan u$. Then $dx = \frac{10}{3}\sec^2 u\,du$.

$$I = \int \frac{900000}{(100+9x^2)^3}\,dx = 3\int \cos^4 u\,du$$
$$= 3\int\left(\frac{1}{8}\cos 4u + \frac{1}{2}\cos 2u + \frac{3}{8}\right)du$$
$$= \frac{3}{32}\sin 4u + \frac{3}{4}\sin 2u + \frac{9}{8}u + c$$
$$= \frac{3}{4}\sin u\cos^3 u + \frac{9}{8}\sin u\cos u + \frac{9}{8}u + c.$$

If $\tan u = \frac{3x}{10}$, then $\sin u = \frac{3x}{\sqrt{100+9x^2}}$ and $\cos u = \frac{10}{\sqrt{100+9x^2}}$. Thus

$$I = \frac{2250x}{(100+9x^2)^2} + \frac{135x}{4(100+9x^2)} + \frac{9}{8}\arctan\left(\frac{3x}{10}\right) + c$$
$$= \frac{22500x + 1215x^3}{4(100+9x^2)^2} + \frac{9}{8}\arctan\left(\frac{3x}{10}\right) + c.$$

So $x = \frac{10}{3}\tan u$ gives the stated integral.

3 j)

Let $49x^2 = 25\tanh^2 u$, so $7x = 5\tanh u$ and $x = \frac{5}{7}\tanh u$. Then $dx = \frac{5}{7}\operatorname{sech}^2 u\,du$.

$$I = \int \frac{280}{(25-49x^2)^2}\,dx = \frac{8}{25}\int \cosh^2 u\,du = \frac{2}{25}\sinh 2u + \frac{4}{25}u + c$$
$$= \frac{4}{25}\sinh u\cosh u + \frac{4}{25}u + c.$$

If $\tanh u = \frac{7}{5}x$, then $\cosh u = \frac{5}{\sqrt{25-49x^2}}$ and $\sinh u = \frac{7x}{\sqrt{25-49x^2}}$. Hence

$$I = \frac{28x}{5(25-49x^2)} + \frac{4}{25}\operatorname{artanh}\left(\frac{7x}{5}\right) + c.$$

So $x = \frac{5}{7}\tanh u$ gives the stated integral.

3 k)

Let $2x^2 = 25\tan^2 u$, so $\sqrt{2}x = 5\tan u$ and $x = \frac{5}{\sqrt{2}}\tan u$. Then $dx = \frac{5\sqrt{2}}{2}\sec^2 u\,du$.

$$I = \int \frac{250}{(25+2x^2)^2}\,dx = \sqrt{2}\int \cos^2 u\,du = \frac{\sqrt{2}}{4}\sin 2u + \frac{\sqrt{2}}{2}u + c$$
$$= \frac{\sqrt{2}}{2}\sin u\cos u + \frac{\sqrt{2}}{2}u + c.$$

If $\tan u = \frac{\sqrt{2}x}{5}$, then $\sin u = \frac{\sqrt{2}x}{\sqrt{25+2x^2}}$ and $\cos u = \frac{5}{\sqrt{25+2x^2}}$. Therefore

$$I = \frac{5x}{25+2x^2} + \frac{\sqrt{2}}{2}\arctan\left(\frac{\sqrt{2}x}{5}\right) + c.$$

So $x = \frac{5}{\sqrt{2}}\tan u$ gives the stated integral.

3 l)

Let $3x^2 = 4\cosh^2 u$, so $\sqrt{3}x = 2\cosh u$ and $x = \frac{2}{\sqrt{3}}\cosh u$. Then $dx = \frac{2\sqrt{3}}{3}\sinh u\,du$.

$$I = \int (3x^2-4)^{3/2}\,dx = \frac{16\sqrt{3}}{3}\int \sinh^4 u\,du.$$

Using $\sinh^2 A = \frac{1}{2}\cosh 2A - \frac{1}{2}$,

$$\begin{aligned}\sinh^4 u &= \frac{1}{8}\cosh 4u - \frac{1}{2}\cosh 2u + \frac{3}{8},\\ I &= \frac{\sqrt{3}}{6}\sinh 4u - \frac{4\sqrt{3}}{3}\sinh 2u + 2\sqrt{3}u + c\\ &= -2\sqrt{3}\sinh u\cosh u + \frac{4\sqrt{3}}{3}\sinh^3 u\cosh u + 2\sqrt{3}u + c\\ &= \frac{1}{4}x(3x^2-10)\sqrt{3x^2-4} + 2\sqrt{3}\,\mathrm{arcosh}\left(\frac{\sqrt{3}x}{2}\right) + c.\end{aligned}$$

So $x = \frac{2}{\sqrt{3}}\cosh u$ gives the stated integral.

4 a)

Let $(3x-1)^2 = \sin^2 u$, so $3x-1 = \sin u$ and $x = \frac{1}{3} + \frac{1}{3}\sin u$. Then $dx = \frac{1}{3}\cos u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{1-(3x-1)^2}}\,dx &= \frac{1}{3}\int \frac{\cos u}{\sqrt{\cos^2 u}}\,du = \frac{1}{3}\int 1\,du\\ &= \frac{1}{3}u + c = \frac{1}{3}\arcsin(3x-1) + c.\end{aligned}$$

So $x = \frac{1}{3} + \frac{1}{3}\sin u$ gives the stated integral.

4 b)

Let $(x-1)^2 = 9\sinh^2 u$, so $x = 1 + 3\sinh u$ and $dx = 3\cosh u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{(x-1)^2+9}}\,dx &= \int \frac{3\cosh u}{\sqrt{9(\sinh^2 u+1)}}\,du = \int \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du\\ &= \int 1\,du = u + c = \mathrm{arsinh}\left(\frac{x-1}{3}\right) + c.\end{aligned}$$

So $x = 1 + 3\sinh u$ gives the stated integral.

4 c)

Let $(2x+5)^2 = 3\sin^2 u$, so $x = -\frac{5}{2} + \frac{\sqrt{3}}{2}\sin u$ and $dx = \frac{\sqrt{3}}{2}\cos u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{3-(2x+5)^2}}\,dx &= \frac{1}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\,du = \frac{1}{2}\int 1\,du\\ &= \frac{1}{2}u + c = \frac{1}{2}\arcsin\left(\frac{2x+5}{\sqrt{3}}\right) + c.\end{aligned}$$

So $x = -\frac{5}{2} + \frac{\sqrt{3}}{2}\sin u$ gives the stated integral.

4 d)

Let $(3x-2)^2 = 8\sinh^2 u$, so $x = \frac{2}{3} + \frac{2\sqrt{2}}{3}\sinh u$ and $dx = \frac{2\sqrt{2}}{3}\cosh u\, du$.

$$\begin{aligned}\int \frac{1}{\sqrt{(3x-2)^2+8}}\, dx &= \frac{1}{3}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\, du = \frac{1}{3}\int 1\, du \\ &= \frac{1}{3}u + c = \frac{1}{3}\operatorname{arsinh}\left(\frac{3x-2}{2\sqrt{2}}\right) + c.\end{aligned}$$

So $x = \frac{2}{3} + \frac{2\sqrt{2}}{3}\sinh u$ gives the stated integral.

4 e)

Let $(4x-1)^2 = 8\tan^2 u$, so $x = \frac{1}{4} + \frac{\sqrt{2}}{2}\tan u$ and $dx = \frac{\sqrt{2}}{2}\sec^2 u\, du$.

$$\begin{aligned}\int \frac{1}{8+(4x-1)^2}\, dx &= \frac{\sqrt{2}}{16}\int \frac{\sec^2 u}{\sec^2 u}\, du = \frac{\sqrt{2}}{16}u + c \\ &= \frac{\sqrt{2}}{16}\arctan\left(\frac{4x-1}{2\sqrt{2}}\right) + c.\end{aligned}$$

So $x = \frac{1}{4} + \frac{\sqrt{2}}{2}\tan u$ gives the stated integral.

4 f)

Let $(5x+3)^2 = 11\tanh^2 u$, so $x = -\frac{3}{5} + \frac{\sqrt{11}}{5}\tanh u$ and $dx = \frac{\sqrt{11}}{5}\operatorname{sech}^2 u\, du$.

$$\begin{aligned}\int \frac{4}{11-(5x+3)^2}\, dx &= \frac{4\sqrt{11}}{55}\int \frac{\operatorname{sech}^2 u}{\operatorname{sech}^2 u}\, du = \frac{4\sqrt{11}}{55}u + c \\ &= \frac{4\sqrt{11}}{55}\operatorname{artanh}\left(\frac{5x+3}{\sqrt{11}}\right) + c.\end{aligned}$$

So $x = -\frac{3}{5} + \frac{\sqrt{11}}{5}\tanh u$ gives the stated integral.

5 a)

First, $4x^2 + 12x + 10 = 4\left(x+\frac{3}{2}\right)^2 + 1$. Let $4\left(x+\frac{3}{2}\right)^2 = \tan^2 u$, so $x = -\frac{3}{2} + \frac{1}{2}\tan u$ and $dx = \frac{1}{2}\sec^2 u\, du$.

$$\begin{aligned}\int \frac{1}{4x^2+12x+10}\, dx &= \frac{1}{2}\int \frac{\sec^2 u}{\tan^2 u + 1}\, du = \frac{1}{2}\int 1\, du \\ &= \frac{1}{2}u + c = \frac{1}{2}\arctan(2x+3) + c.\end{aligned}$$

So $x = -\frac{3}{2} + \frac{1}{2}\tan u$ gives the stated integral.

5 b)

First, $3 - 4x - 4x^2 = 4 - 4\left(x+\frac{1}{2}\right)^2$. Let $4\left(x+\frac{1}{2}\right)^2 = 4\sin^2 u$, so $x = -\frac{1}{2} + \sin u$ and $dx = \cos u\, du$.

$$\begin{aligned}\int \frac{3}{\sqrt{3-4x-4x^2}}\, dx &= \frac{3}{2}\int \frac{\cos u}{\sqrt{\cos^2 u}}\, du = \frac{3}{2}\int 1\, du \\ &= \frac{3}{2}u + c = \frac{3}{2}\arcsin\left(x+\frac{1}{2}\right) + c.\end{aligned}$$

So $x = -\frac{1}{2} + \sin u$ gives the stated integral.

5 c)

First, $x^2-6x+14=(x-3)^2+5$. Let $(x-3)^2=5\tan^2 u$, so $x=3+\sqrt{5}\tan u$ and $dx=\sqrt{5}\sec^2 u\,du$.

$$\begin{aligned}\int \frac{2}{x^2-6x+14}\,dx &= \frac{2\sqrt{5}}{5}\int \frac{\sec^2 u}{\tan^2 u+1}\,du = \frac{2\sqrt{5}}{5}u+c\\ &= \frac{2\sqrt{5}}{5}\arctan\left(\frac{x-3}{\sqrt{5}}\right)+c.\end{aligned}$$

So $x=3+\sqrt{5}\tan u$ gives the stated integral.

5 d)

First, $7+4x-x^2=11-(x-2)^2$. Let $(x-2)^2=11\sin^2 u$, so $x=2+\sqrt{11}\sin u$ and $dx=\sqrt{11}\cos u\,du$.

$$\begin{aligned}\int \frac{1}{\sqrt{7+4x-x^2}}\,dx &= \int \frac{\cos u}{\sqrt{\cos^2 u}}\,du = \int 1\,du\\ &= u+c = \arcsin\left(\frac{x-2}{\sqrt{11}}\right)+c.\end{aligned}$$

So $x=2+\sqrt{11}\sin u$ gives the stated integral.

5 e)

First, $3x^2+8x+6=3\left(x+\frac{4}{3}\right)^2+\frac{2}{3}$. Let $3\left(x+\frac{4}{3}\right)^2=\frac{2}{3}\tan^2 u$, so $x=-\frac{4}{3}+\frac{\sqrt{2}}{3}\tan u$ and $dx=\frac{\sqrt{2}}{3}\sec^2 u\,du$.

$$\begin{aligned}\int \frac{4}{3x^2+8x+6}\,dx &= 2\sqrt{2}\int \frac{\sec^2 u}{\tan^2 u+1}\,du = 2\sqrt{2}u+c\\ &= 2\sqrt{2}\arctan\left(\frac{3x+4}{\sqrt{2}}\right)+c.\end{aligned}$$

So $x=-\frac{4}{3}+\frac{\sqrt{2}}{3}\tan u$ gives the stated integral.

5 f)

First, $5x^2+2x+12=5\left(x+\frac{1}{5}\right)^2+\frac{59}{5}$. Let $5\left(x+\frac{1}{5}\right)^2=\frac{59}{5}\sinh^2 u$, so $x=-\frac{1}{5}+\frac{\sqrt{59}}{5}\sinh u$ and $dx=\frac{\sqrt{59}}{5}\cosh u\,du$.

$$\begin{aligned}\int \frac{10}{\sqrt{5x^2+2x+12}}\,dx &= 2\sqrt{5}\int \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du = 2\sqrt{5}u+c\\ &= 2\sqrt{5}\operatorname{arsinh}\left(\frac{5x+1}{\sqrt{59}}\right)+c.\end{aligned}$$

So $x=-\frac{1}{5}+\frac{\sqrt{59}}{5}\sinh u$ gives the stated integral.

6 a)

Let $(2x-1)^2=5\sin^2 u$, so $x=\frac{1}{2}+\frac{\sqrt{5}}{2}\sin u$ and $dx=\frac{\sqrt{5}}{2}\cos u\,du$. When $x=0$, $u=\arcsin(-\sqrt{5}/5)$; when $x=1$, $u=\arcsin(\sqrt{5}/5)$.

$$\begin{aligned}\int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\,dx &= \int_{\arcsin(-\sqrt{5}/5)}^{\arcsin(\sqrt{5}/5)} \frac{\cos u}{\sqrt{\cos^2 u}}\,du\\ &= [u]_{\arcsin(-\sqrt{5}/5)}^{\arcsin(\sqrt{5}/5)}\end{aligned}$$

$$= \arcsin\left(\frac{\sqrt{5}}{5}\right) - \arcsin\left(-\frac{\sqrt{5}}{5}\right)$$
$$= 2\arcsin\left(\frac{\sqrt{5}}{5}\right).$$

So $x = \frac{1}{2} + \frac{\sqrt{5}}{2}\sin u$ gives the stated result.

6 b)

Let $(x-1)^2 = 4\sinh^2 u$, so $x = 1 + 2\sinh u$ and $dx = 2\cosh u\, du$. When $x = 0$, $u = -\operatorname{arsinh}(1/2)$; when $x = 2$, $u = \operatorname{arsinh}(1/2)$.

$$\int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\,dx = 8\int_{-\operatorname{arsinh}(1/2)}^{\operatorname{arsinh}(1/2)} \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du = [8u]_{-\operatorname{arsinh}(1/2)}^{\operatorname{arsinh}(1/2)}$$
$$= 16\operatorname{arsinh}\left(\frac{1}{2}\right) = 16\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)$$
$$= 8\ln\left(\frac{3+\sqrt{5}}{2}\right).$$

So $x = 1 + 2\sinh u$ gives the stated result.

6 c)

First, $45+24x+4x^2 = 9+4(x+3)^2$. Let $4(x+3)^2 = 9\tan^2 u$, so $x = -3+\frac{3}{2}\tan u$ and $dx = \frac{3}{2}\sec^2 u\, du$. When $x = -3$, $u = 0$; when $x = -\frac{3}{2}$, $u = \frac{\pi}{4}$.

$$\int_{-3}^{-3/2} \frac{3}{45+24x+4x^2}\,dx = \int_0^{\pi/4} \frac{3}{9+9\tan^2 u}\cdot\frac{3}{2}\sec^2 u\,du$$
$$= \frac{1}{2}\int_0^{\pi/4} 1\,du = \left[\frac{1}{2}u\right]_0^{\pi/4} = \frac{\pi}{8}.$$

So $x = -3 + \frac{3}{2}\tan u$ gives the stated result.

6 d)

First, $4x^2 - 12x + 15 = 6 + 4\left(x-\frac{3}{2}\right)^2$. Let $4\left(x-\frac{3}{2}\right)^2 = 6\sinh^2 u$, so $x = \frac{3}{2} + \frac{\sqrt{6}}{2}\sinh u$ and $dx = \frac{\sqrt{6}}{2}\cosh u\, du$. When $x = 0$, $u = \operatorname{arsinh}(-\sqrt{3}/\sqrt{2})$; when $x = \frac{3}{2}$, $u = 0$.

$$\int_0^{3/2} \frac{2}{\sqrt{4x^2-12x+15}}\,dx = \int_{\operatorname{arsinh}(-\sqrt{3}/\sqrt{2})}^0 \frac{\cosh u}{\sqrt{\cosh^2 u}}\,du = [u]_{\operatorname{arsinh}(-\sqrt{3}/\sqrt{2})}^0$$
$$= -\operatorname{arsinh}\left(-\frac{\sqrt{3}}{\sqrt{2}}\right) = -\ln\left(-\frac{\sqrt{3}}{\sqrt{2}} + \sqrt{\frac{3}{2}+1}\right)$$
$$= -\ln\left(\frac{-\sqrt{3}+\sqrt{5}}{\sqrt{2}}\right) = \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right).$$

So $x = \frac{3}{2} + \frac{\sqrt{6}}{2}\sinh u$ gives the stated result.

6 e)

First, $x^2+2x+10 = 9+(x+1)^2$. Let $(x+1)^2 = 9\sinh^2 u$, so $x = -1+3\sinh u$ and $dx = 3\cosh u\, du$. When $x = -1$, $u = 0$; when $x = \frac{5}{4}$, $u = \operatorname{arsinh}(3/4) = \ln 2$.

$$I = \int_{-1}^{5/4} \sqrt{x^2+2x+10}\,dx = 9\int_0^{\ln 2} \cosh^2 u\,du$$

$$= 9\left[\frac{1}{4}\sinh 2u + \frac{1}{2}u\right]_0^{\ln 2}$$
$$= \frac{9}{4}\sinh(\ln 4) + \frac{9}{2}\ln 2$$
$$= \frac{9}{8}\left(4 - \frac{1}{4}\right) + \frac{9}{2}\ln 2 = \frac{135}{32} + \frac{9}{2}\ln 2.$$

So $x = -1 + 3\sinh u$ gives the stated result.

7)

First, $3 + 2x - x^2 = 4 - (x-1)^2$. Let $(x-1)^2 = 4\sin^2 u$, so $x = 1 + 2\sin u$ and $dx = 2\cos u\, du$.

$$\int \frac{5}{\sqrt{3+2x-x^2}}\, dx = 5\int \frac{\cos u}{\sqrt{\cos^2 u}}\, du = 5u + c$$
$$= 5\arcsin\left(\frac{x-1}{2}\right) + c.$$

So that

$$\int_0^1 \frac{5}{\sqrt{3+2x-x^2}}\, dx = \left[5\arcsin\left(\frac{x-1}{2}\right)\right]_0^1$$
$$= 5\arcsin 0 - 5\arcsin\left(-\frac{1}{2}\right) = \frac{5\pi}{6}\ \text{units}^2.$$

8)

First, $6+4x+x^2 = (x+2)^2+2$. Let $(x+2)^2 = 2\tan^2 u$, so $x = -2+\sqrt{2}\tan u$ and $dx = \sqrt{2}\sec^2 u\, du$.

$$\int \frac{4}{6+4x+x^2}\, dx = 2\sqrt{2}\int \frac{\sec^2 u}{\tan^2 u + 1}\, du = 2\sqrt{2}u + c$$
$$= 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right) + c.$$

So $\displaystyle\int \frac{4}{6+4x+x^2}\, dx = 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right) + c$. Then

$$\int_{\sqrt{2}-2}^{\sqrt{6}-2}\left(x + \frac{4}{6+4x+x^2}\right) dx = \left[\frac{1}{2}x^2 + 2\sqrt{2}\arctan\left(\frac{x+2}{\sqrt{2}}\right)\right]_{\sqrt{2}-2}^{\sqrt{6}-2}$$
$$= \left(5 - 2\sqrt{6} + 2\sqrt{2}\arctan\sqrt{3}\right) - \left(3 - 2\sqrt{2} + 2\sqrt{2}\arctan 1\right)$$
$$= 2 + 2\sqrt{2} - 2\sqrt{6} + \left(\frac{2}{3}\sqrt{2} - \frac{1}{2}\sqrt{2}\right)\pi$$
$$= \left(2 + 2\sqrt{2} - 2\sqrt{6} + \frac{\sqrt{2}}{6}\pi\right)\ \text{units}^2.$$